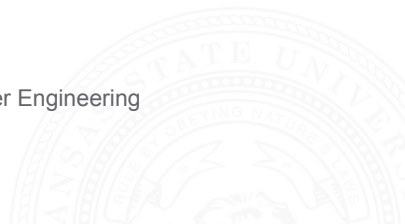


Transformer Modeling I

Ideal voltage, current, and impedance relationships

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Lecture outline

- 1 Explain how AC excitation produces voltage transformation
- 2 Identify winding variables and apply four ideal assumptions
- 3 Derive voltage, current, power, and impedance relationships

A physical picture establishes the variables. The ideal assumptions then support one relationship at a time.

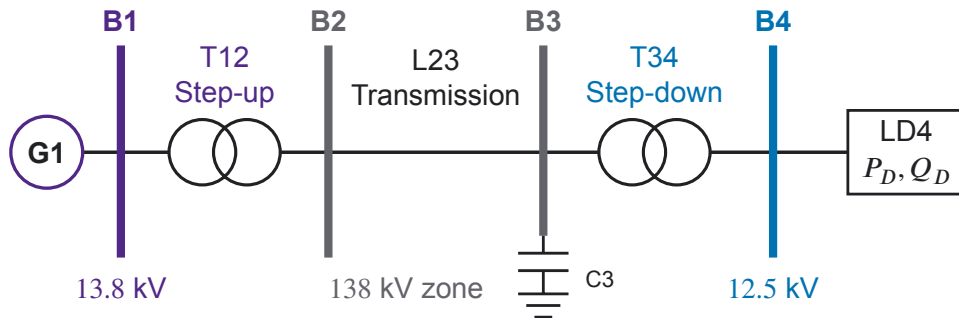
PART 1

Physical operation and variables

Transformer operation, winding quantities, and the magnetic core

Transformers in the power system

SYMBOLIC REPRESENTATION: GENERATION, TRANSMISSION, AND LOAD



Step up for high-voltage transmission; step down to supply lower-voltage loads.

Transformers in Power Systems



Photo: Yoshieslunchbox / Wikimedia Commons, CC BY-SA 4.0.

138/12.5 kV

Substation power transformer

Bushings (top)

Insulated connections to the windings.

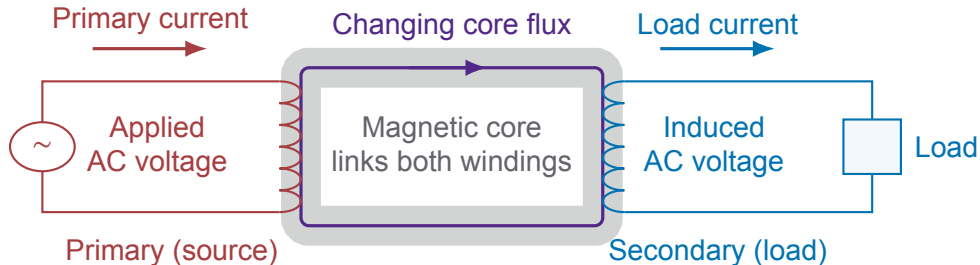
Main tank

Contains the core, windings, and insulating oil.

Radiators (right)

Remove heat from the oil.

How a transformer works



1. AC excitation

Applied voltage establishes alternating flux in the core.

2. Induced voltage

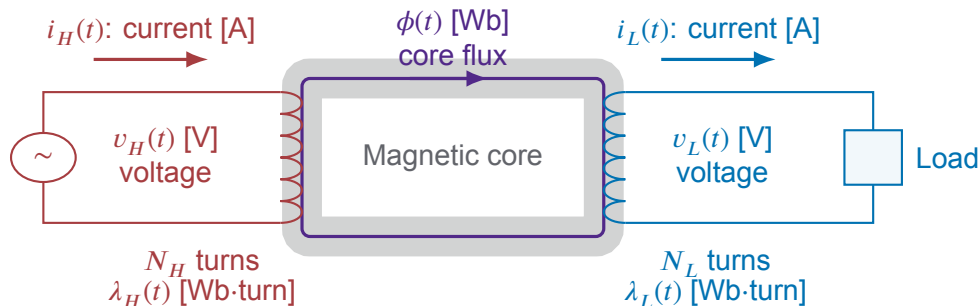
Changing flux induces voltage. More turns give more voltage.

3. Load response

A load draws secondary current and additional primary current.

Secondary voltage exists at no load; secondary current requires a connected load.

Winding variables and units



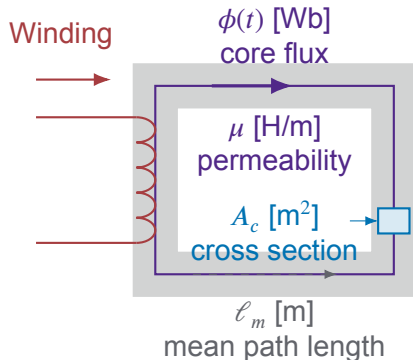
λ : flux linkage (sum over all turns).

N : turn count [dimensionless]; t : time [s].

H/L: higher/lower-voltage windings.

Here: H is primary; L is secondary.

Core material and geometry



Magnetic reluctance

$$\mathcal{R} = \frac{\ell_m}{\mu A_c}$$

$\mathcal{R}$: opposition to flux
[A·turn/Wb]

Higher permeability μ means lower reluctance.

Permeability is a material property; current does not create it.

Magnetizing current

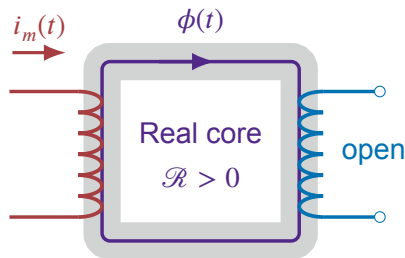
Current needed to establish core flux:

$$N_H i_m(t) = \mathcal{R} \phi(t)$$

$i_m(t)$: magnetizing current [A]

At no load, $i_L = 0$.

A real core still needs i_m .



Magnetizing current is one component of no-load current; core loss adds another.

PART 2

Ideal transformer model

Four assumptions and the voltage, current, power, and impedance relationships

Zero winding resistance

Ideal assumption 1

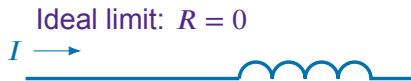
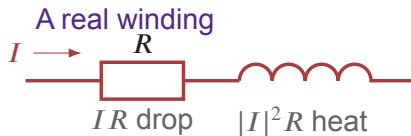
Winding resistance [Ω]

$$R_H = R_L = 0$$

No resistive voltage drop.

No winding heat loss:

$$P_{\text{cu}} = 0 \quad [\text{W}]$$



Current flows with no copper loss.

Zero leakage flux

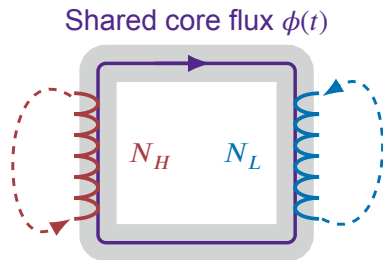
Ideal assumption 2

Leakage flux [Wb]

$$\phi_{\ell H} = \phi_{\ell L} = 0$$

Every turn links the same core flux $\phi(t)$.

No leakage-inductance voltage drop.



Dashed: real leakage paths
Ideal model omits these paths.

Infinite core permeability

Ideal assumption 3

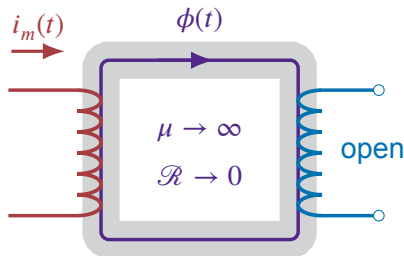
Infinite permeability

$$\mu \rightarrow \infty \Rightarrow \mathcal{R} \rightarrow 0$$

For a finite core flux:

$$i_m(t) \rightarrow 0$$

Loaded winding currents can still flow.



Magnetizing component only

No real power dissipated in the core:

$$P_{\text{core}} = 0 \quad [\text{W}]$$

Hysteresis: cyclic magnetization.

Eddy currents: circulation in steel.

Ideal core-loss current is zero.



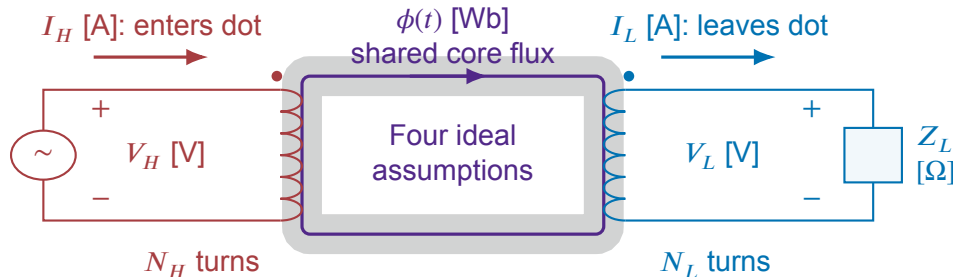
B : flux density. H_m : field strength.

Eddy-current loss



Changing flux through the cross section

Ideal model and reference directions



Uppercase V, I : RMS phasors. Voltages are positive at the dotted terminals.

$a \equiv N_H/N_L > 1$: turns ratio [dimensionless]. Z_L : load impedance [Ω].

Flux linkage in an ideal winding

Each of the N turns links the same flux $\phi(t)$.

$$\lambda(t) = \underbrace{\phi(t) + \phi(t) + \cdots + \phi(t)}_{N \text{ turns}}$$

$$\lambda(t) = N\phi(t)$$

ϕ : flux [Wb]. N : turns [dimensionless]. λ : flux linkage [Wb·turn].

Equal flux per turn

$$\lambda_H = N_H \phi, \quad \lambda_L = N_L \phi$$

$$\boxed{\frac{\lambda_H}{N_H} = \frac{\lambda_L}{N_L} = \phi}$$

$$\lambda_H = a \lambda_L$$

Same flux through each turn

$\phi(t)$ [Wb]

N_H turns
 $\lambda_H = N_H \phi$

N_L turns
 $\lambda_L = N_L \phi$

Sum over each winding
to obtain its flux linkage

The windings share flux per turn. Their total flux linkages scale with turn count.

Induced winding voltage

Faraday's law

$$e(t) = \frac{d\lambda(t)}{dt}$$

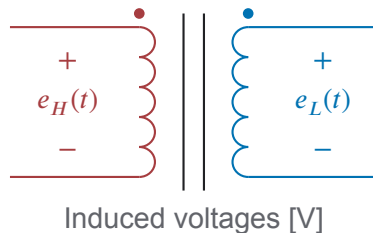
$$e_H(t) = N_H \frac{d\phi(t)}{dt}$$

$$e_L(t) = N_L \frac{d\phi(t)}{dt}$$

e_H, e_L : induced voltages [V].

$d\phi/dt$: flux rate [Wb/s = V].

Both windings link
the same changing flux



Changing flux induces voltage in both windings.

Voltage ratio

$$\frac{e_H(t)}{N_H} = \frac{e_L(t)}{N_L}$$

No winding drops: $v_H = e_H$, $v_L = e_L$.

$$v_H(t) = a v_L(t), \quad a = \frac{N_H}{N_L}$$

$$\boxed{V_H = a V_L}$$



Step-up operation



Step-down operation

More turns give higher voltage. The declared voltage phasors have the same angle.

Ampere-turn balance

$$N_H i_H(t) - N_L i_L(t) = \mathcal{R} \phi(t)$$

Ni : magnetomotive force [A·turn].

$$\mathcal{R} = 0 \quad \Rightarrow \quad \boxed{N_H i_H(t) = N_L i_L(t)}$$

Primary ampere-turns



$$N_H i_H(t)$$

$$N_L i_L(t)$$



Secondary ampere-turns

Equal and opposing
in the ideal core

The declared current arrows produce equal and opposing ampere-turns.

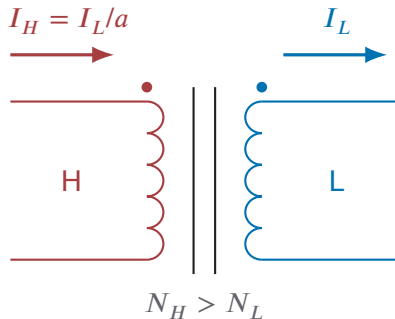
Current ratio

$$N_H I_H = N_L I_L$$

$$I_H = \frac{N_L}{N_H} I_L$$

$$I_H = \frac{I_L}{a}$$

I_H, I_L : RMS current phasors [A].
Same phase angle with these reference
arrows.



The higher-voltage winding carries the smaller current.

Power quantities and units

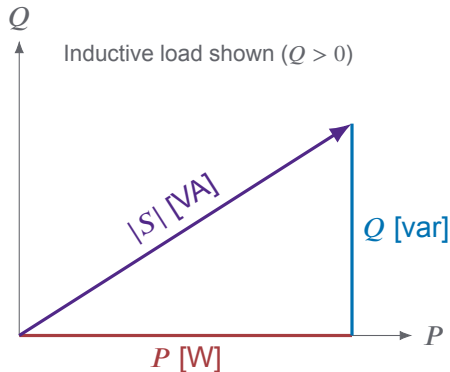
$$S = P + jQ$$

S Complex power VA

P Real power W

Q Reactive power var

$|S|$ Apparent power VA



Power factor $\text{pf} = P/|S|$ [dimensionless].

Complex-power conservation

For one phase, using RMS phasors:

$$S = VI^*$$

$$S_H = (aV_L) \left(\frac{I_L}{a} \right)^* = V_L I_L^*$$

$$S_H = S_L$$

*: complex conjugate. S_H : input power. S_L : power delivered to the load.

Real power, reactive power, apparent power, and power factor are unchanged.

Load impedance referred to H

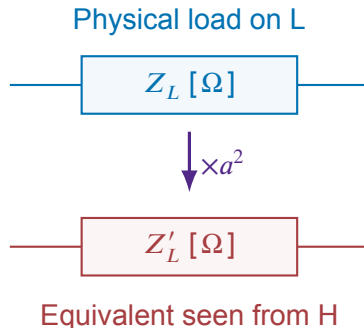
$$Z = \frac{V}{I}$$

$$Z'_L \equiv \frac{V_H}{I_H} = \frac{aV_L}{I_L/a}$$

$$Z'_L = a^2 Z_L$$

$Z_L = V_L/I_L$: physical L-side load [Ω].

Z'_L : its equivalent seen from H [Ω].



One factor of a comes from voltage; the other comes from the inverse current ratio.

Impedance referred to L

$$Z_H = a^2 Z'_H$$

$$Z'_H = \frac{Z_H}{a^2}$$

Z_H : physical H-side impedance [Ω].

Z'_H : its equivalent seen from L [Ω].

Physical impedance on H



Equivalent seen from L

Keep $a = N_H/N_L$ fixed. A prime marks the equivalent referred impedance.

Low-voltage-side equivalent circuit

The L winding supplies load Z_L .

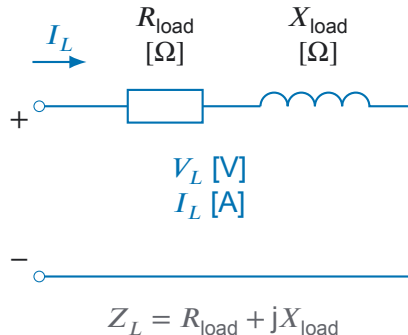
$$Z_L = \frac{V_L}{I_L}$$

$$= R_{\text{load}} + jX_{\text{load}}$$

R_{load} : load resistance $[\Omega]$.

X_{load} : load reactance $[\Omega]$.

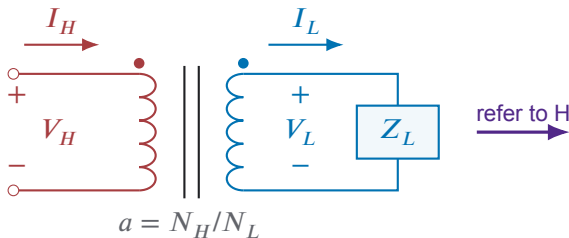
Inductive load: $X_{\text{load}} > 0$.



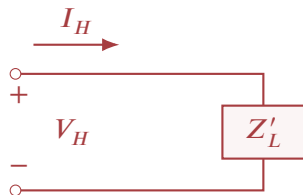
These elements model the load connected to the L winding.

Equivalent circuit referred to H

Transformer and load



Equivalent seen from H



One circuit on the H side

$$Z'_L = \frac{V_H}{I_H} = a^2 Z_L \quad [\Omega]$$

For $Z_L = R_{\text{load}} + jX_{\text{load}}$, scale both resistance and reactance by a^2 .

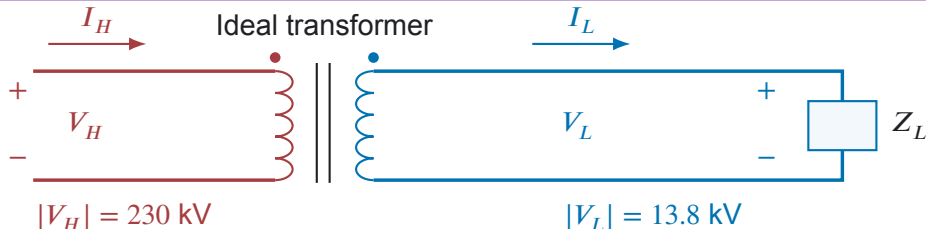
The same applied V_H produces the same input I_H in both circuits.

PART 3

Worked example

A single-phase 230/13.8-kV transformer supplying a 60-MVA load

Example setup



Single-phase transformer. Dots define voltage polarity; magnitudes are RMS.

Given load

$|S_L| = 60 \text{ MVA}$, $\text{pf} = 0.80$ lagging.
Single-phase load supplied at 13.8-kV
RMS.

Find and verify

a , I_L , I_H , physical Z_L , and referred Z'_L .
Check apparent power and direct
high-side ohms.

Use the same single-phase model and reference directions as in the theory.

Example: Step 1

Complex load power

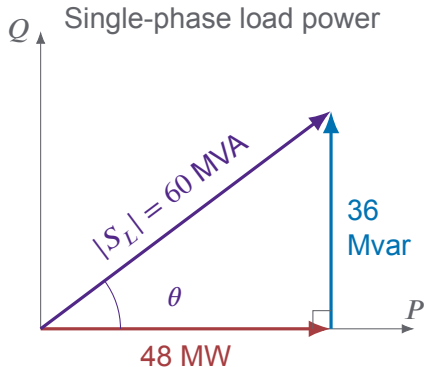
$$\theta = \cos^{-1}(0.80) = 36.870^\circ$$

$$P_L = 60(0.80) = 48 \text{ MW}$$

$$Q_L = 60(0.60) = 36 \text{ Mvar}$$

$$S_L = 48 + j36 \text{ MVA}$$

θ : power-factor angle. Positive Q_L indicates an inductive load.



The positive power-factor angle means current lags the load voltage by 36.870° .

Example: Step 2

Turns ratio and winding voltages

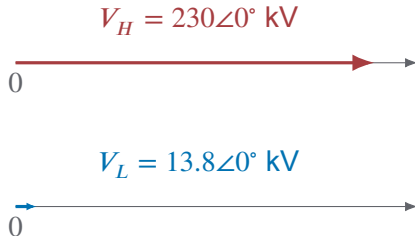
$$a = \frac{N_H}{N_L} = \frac{230}{13.8} \approx \boxed{16.6667}$$

$$V_L = 13.8\angle 0^\circ \text{ kV}$$

$$V_H = aV_L = 230\angle 0^\circ \text{ kV}$$

Choose V_L as the reference. Both voltages are RMS phasors.

Winding-voltage phasors



Same voltage scale; same phase angle

The turns ratio scales the winding voltage directly: $V_H = aV_L$.

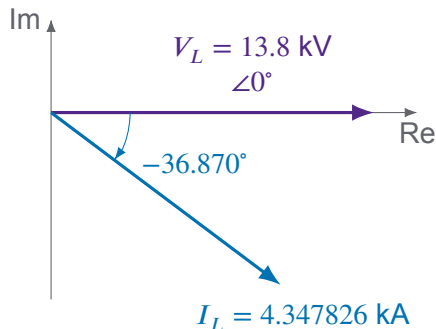
Example: Step 3

Take V_L as the zero-degree reference:

$$S_L^* = 60 \angle (-36.870^\circ) \text{ MVA}$$

$$\begin{aligned} I_L &= \frac{S_L^*}{V_L^*} \\ &= \frac{60 \angle (-36.870^\circ) \text{ MVA}}{13.8 \angle 0^\circ \text{ kV}} \\ &= \boxed{4347.826 \angle (-36.870^\circ) \text{ A}} \end{aligned}$$

Voltage and current phasors



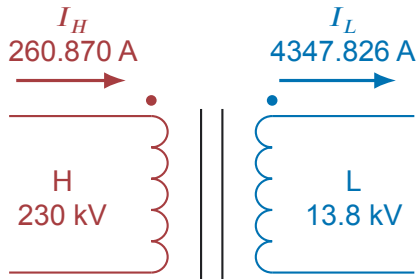
Voltage and current use separate magnitude scales.

Conjugation changes the power angle to $-\theta$; polar division subtracts the voltage angle.

Example: Step 4

Current referred to H

$$\begin{aligned} I_H &= \frac{I_L}{a} \\ &= \frac{4347.826 \angle (-36.870^\circ)}{230/13.8} \\ &= \boxed{260.870 \angle (-36.870^\circ) \text{ A}} \\ \frac{|I_L|}{|I_H|} &\approx 16.6667, \quad \angle I_H = \angle I_L \end{aligned}$$



$$N_H : N_L = 16.6667 : 1$$

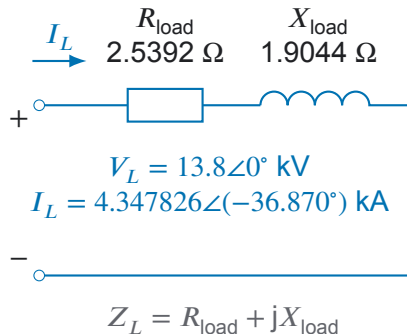
The arrows follow the theory; labels give RMS winding quantities.

The 230-kV current magnitude is 0.0600 times the 13.8-kV current magnitude. The current angle is unchanged.

Example: Step 5

Physical load impedance

$$\begin{aligned} Z_L &= \frac{V_L}{I_L} \\ &= \frac{13.8\angle 0^\circ \text{ kV}}{4.347826\angle(-36.870^\circ) \text{ kA}} \\ &= 3.1740\angle 36.870^\circ \Omega \\ &= \boxed{(2.5392 + j1.9044) \Omega} \end{aligned}$$



Positive reactance and a positive impedance angle agree with the stated lagging power factor.

Example: Step 6

Load referred from L to H

$$a^2 = \left(\frac{230}{13.8} \right)^2 \approx 277.778$$

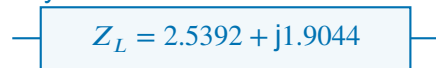
$$Z'_L = a^2 Z_L$$

$$= \left(\frac{230}{13.8} \right)^2 (3.1740 \angle 36.870^\circ) \Omega$$

$$= 881.667 \angle 36.870^\circ \Omega$$

$$= \boxed{(705.333 + j529.000) \Omega}$$

Physical L-side load



Equivalent seen from H



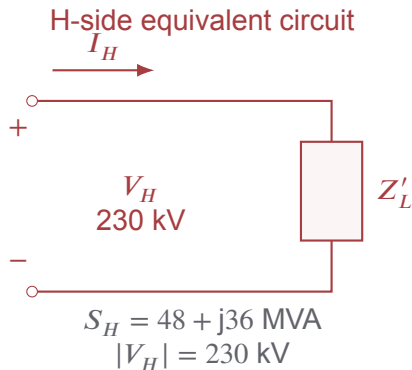
Both labels in Ω

Referral multiplies both resistance and reactance by 277.778 and preserves the impedance angle.

Direct high-side impedance

For this single-phase load, $S_H = V_H I_H^*$:

$$\begin{aligned} Z_{H,\text{direct}} &= \frac{|V_H|^2}{S_H^*} \\ &= \frac{230^2 \angle 0^\circ \text{ kV}^2}{60 \angle (-36.870^\circ) \text{ MVA}} \\ &= 881.667 \angle 36.870^\circ \Omega \\ &= \boxed{(705.333 + j529.000) \Omega} \\ Z_{H,\text{direct}} &= Z'_L \end{aligned}$$



Using H-side voltage and the same load power independently reproduces the referred impedance.

Apparent-power check



$$|S_H| = (230)(0.260870) \\ = 60.000 \text{ MVA}$$

$$|S_L| = (13.8)(4.347826) \\ = 60.000 \text{ MVA}$$

Single-phase check: $|S| = |V||I|$, using RMS winding kV and kA.

$$V_H I_H^* = V_L I_L^* = \boxed{48 + j36 \text{ MVA}}$$

Magnitude and complex-power routes both close across the ideal transformer.

Ideal-transformer summary

- **Ratio convention:** define $a = N_H/N_L = V_H/V_L$ from fixed equipment-side labels, independent of power-flow direction.
- **Current scaling:** ideal current magnitude changes inversely with the turns ratio while the declared polarity fixes phase.
- **Impedance referral:** an impedance referred across the ideal transformer scales by a^2 . Apply the referral once.
- **Power invariance:** an ideal transformer preserves complex power, providing an independent check on voltage and current scaling.

Freeze the H/L convention first. Every voltage, current, and impedance transformation follows from it.